

Fall 2026

**CISC/CMPE 320 : Fundamentals of
Software Development**

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Topic: A Bit of History

Announcements

- Groups will be finalized Tomorrow (Sep 15), hopefully.

Outline

- History of C++
- C++ standards
- The “philosophy” of C++

History of C++

In the beginning there was C

Not really, but let's not go back too far

History of C++: BCPL

- To 1966 specifically
- **BCPL** or “**BasicComputer Programming Language**” designed by Martin Richards at the University of Cambridge.
- Used to write compilers.
- It was the first language that created “O-code” ahead of creating executable code.

History of C++: BCPL

- Before BCPL, In order to run a program on a new platform, the entire compiler program had to be re-written (took 5 man-months...).
- With BCPL, only small part (about 1/5 of the total code) is subject to changes should the platforms changes. That is the part that generated the machine code, the “Virtual Machine.”
- The part that generated the generic O-Code did not have to be changed in this case. Great Idea!
- (Also the first language to use { }, and to use // for comments.)

History of C++: B

- Written at Bell Labs by Ken Thompson with contributions by Dennis Ritchie. First appeared in 1969.
- Stripped down version of BCPL to run on “minicomputers” like the DEC PDP-11.
- Had a single data type, the “word” that was an integer most of the time but could also be a memory reference.
- Gradually evolved into C in the early 70’s.

History of C++: C

- Dennis Ritchie, of Bell Labs, is given credit for C in 1972.
- A procedural language used for systems programming (compilers for example) and for embedded systems.
- Gives low level access to memory and efficient translation to machine code.
- No longer have to code in Assembly language!
- Used on any kind of computer during the late 70's and early 80's including the IBM-PC.
- In 1983 the language was standardized by ANSI, giving birth to "ANSI C", also called "C89".

C with classes

- Was developed by Bjarne Stroustrup at Bell Labs starting in 1979, as an enhancement to C.
- Added to C:
 - classes
 - virtual functions
 - operator overloading
 - multiple inheritance
 - templates
 - exception handling, etc.
- In 1983 the name of the language was changed to C++ (the increment operator...).

C++ Standards

- It took a long time, but the language was first standardized in 1998 by ISO (“International Organization for Standardization”).
- The standard was updated in 2003 and amended in 2005 (310 pages, without the library!).
- Another standard was completed in 2011: “C++11” (August 12, 2011, 1338 pages!).
- C++14 came out on August 18, 2014. A smaller set of changes from C++11.
- C++17 was published in Dec. 2017 (1605 pages).
- And C++20 was released in Dec. 2020.

C++ Standards, Cont.

- But:
 - It takes a while (and a considerable amount of work) to update compilers to the new standards
- for a summary of how various compilers are doing, see:

https://www.google.com/search?q=comparison+between+compilers&sca_esv=e3cd531e34091613&rlz=1C5CHFA_enCA1160CA1160&ei=x_y6aKD_B7GgptQPkojSA&ved=0ahUKEwjg_tWc88GPAXUxkIkEHRLEGakQ4dUDCBA&uact=5&oq=comparison+between+compilers&gs_lp=Egxnd3Mtd2l6LXNlcniAiHGNvbXBhcmIzb24gYmV0d2VlbiBjb21waWxlcniMyBhAAGBYHjIGEAAyFhgeMgYQABgWGB4yBhAAGBYHjIIEAAyFhgeGAoyBhAAGBYHjIIEAAyGAAQYigUYhgMyCxAAGIAEGIoFGIYDMggQABiABBiiBDIFEAAy7wVI4iJQmwIY8RxxAXgBkAEAmAFjoAHECaoBAjE3uAEDyAEA-AEBmAIQoALeCMICChAAGEcY1gQYsAPCAgUQABiABJgDAIgGAZAGCJIHBDE1LjGgB56gAbIHBDE0LjG4B9klwgcEMi0xNsgHQQ&sclient=gws-wiz-serp

C++ Standards, Cont.

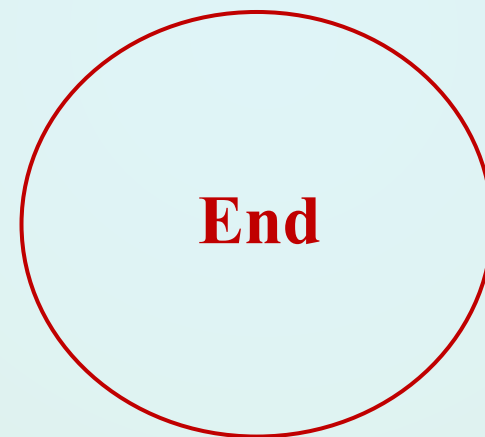
- Compilation is **not** standardized, just the syntax.
- How does Java compare, for example?

The “Philosophy” of C++

- C++ is designed to be a statically typed, general-purpose language that is as efficient and portable as C.
- C++ is designed to directly and comprehensively support multiple programming styles (procedural programming, data abstraction, object-oriented programming, and generic programming).
- *C++ is designed to give the programmer a choice, even if this makes it possible for the programmer to choose incorrectly.*

The “Philosophy” of C++

- C++ is designed to be as compatible with C as possible, therefore providing a smooth transition from C.
- C++ avoids features that are platform specific or not general purpose.
- C++ is designed to function without a sophisticated programming environment.



Next: Basics

